Effect Of Russia-Ukraine War on World Energy Companies in Stock Markets Using Principal Component Analysis

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Declaration

It is hereby declared that

1. The thesis submitted is my/our original work while completing my degree at Noakahli Science and Technology University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material that has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Approval

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Abstract

This research presents a comprehensive exploration of the intricate dynamics of global energy markets, spanning the critical period from July 2018 to June 2023. Employing advanced analytical tools such as Principal Component Analysis (PCA), the study focuses on 50 energy companies, intending initially to scrutinize the impact of the Russia-Ukraine conflict. However, the narrative takes an unexpected turn as the dominating influence of the COVID-19 pandemic emerges during the investigation. Through a meticulous examination of eigenvalue evolution, crisis-state identification heat-maps, continental-based responses, and cluster-based categorization, the study unravels the complex responses of energy corporations to unprecedented global events.

The exploration of eigenvalues unravels a stable market environment abruptly disrupted by the onset of the COVID-19 pandemic. The crisis-state heat-map vividly illustrates the challenges faced by energy companies during this turbulent period, prioritizing the pandemic's impact over geopolitical issues. Cluster-based categorization becomes a focal point, revealing the strategic repositioning of energy companies. During the COVID-19 pandemic, we identified four distinct clusters by the WARD method, each representing companies from different continents. But in the COVID-19 crisis, a big cluster was created. The significant transition and increased PC1 values for a majority of companies during the crisis indicate a pronounced upheaval. In contrast, the Russia-Ukraine war's impact, though maintaining four clusters, exhibits less pronounced transitions. Again, Continental responses, depicted in the R Evolution graph, underscore the resilience of global markets, with a notable shape transition during the COVID-19 pandemic and European companies show a significant changes during war's crisis.

The research not only captures the trans-formative effect of the COVID-19 crisis but also sheds light on the interconnected of global markets and the adaptive strategies adopted by energy companies. Despite the unforeseen pivot in focus, the study contributes valuable insights into the nuanced and resilient nature of energy markets in the face of multifaceted challenges. Acknowledging its limitations, the research also provides a road-map for future investigations in this dynamic field. This work offers a holistic understanding of the intricate interactions between geopolitical events and energy markets, emphasizing the unexpected dominance of the COVID-19 pandemic in shaping the trajectory of global energy companies.

Keywords: Energy Markets; Global Crises; Principal Component Analysis; COVID-19 Impact; Russia-Ukraine Conflict;

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Abbreviations

CAARs - Cumulative Average Abnormal Returns

COVID − 19 - Coronavirus Disease 2019

EMH - Efficient Market Hypothesis

LSTM - Long Short-Term Memory

PCA - Principal Component Analysis

Chapter 1

Introduction

1.1 Background

In February 2022, the world witnessed a significant geopolitical event with far-reaching implications—the outbreak of armed conflict between Russia and Ukraine. The conflict not only shook the foundations of international relations but also had profound consequences for various sectors, including the global energy industry. The energy sector, being inherently sensitive to geopolitical dynamics, experienced heightened volatility in the wake of the conflict [1]. The interplay between geopolitical events and financial markets, particularly the stock performances of energy companies, became a subject of paramount interest.
The Russia-Ukraine war unfolded against a backdrop of complex historical, political, and economic factors, contributing to a tense and unpredictable global environment. The energy landscape, characterized by intricate supply chains, geopolitical dependencies, and fluctuating commodity prices, faced unprecedented challenges. The war's potential to disrupt energy supplies, influence market sentiment, and impact the financial health of energy companies prompted a need for in-depth analysis.

1.2 Rationale for the Study

This research seeks to address the gap in understanding the specific impacts of the Russia-Ukraine war on the stock markets of energy companies. Geopolitical events have historically demonstrated the ability to induce market fluctuations, and the energy sector, as a critical component of global commerce, is susceptible to such disruptions [2]. By investigating the repercussions on energy company stock prices, this study aims to unravel the underlying factors contributing to market movements and provide insights that can inform investment decisions, risk management strategies, and policy formulation.

1.3 Objectives of the Study

The primary objectives of this research are:

- To analyze the stock market performance of global energy companies during and in the aftermath of the Russia-Ukraine war.

- To identify key factors influencing the stock prices of energy companies in response to geopolitical tensions.
- To assess the extent to which the conflict impacts different segments of the energy sector.
- To utilize Principal Component Analysis (PCA) as a tool for understanding the multidimensional relationships between geopolitical indicators and stock market performance.

1.4 Research Questions

To achieve the stated objectives, the study addresses the following research questions:

- How did the Russia-Ukraine war impact the stock prices of energy companies globally?
- What are the key geopolitical indicators influencing the stock market performance of energy companies?
- How do different segments within the energy sector respond to geopolitical tensions?
- How can Principal Component Analysis contribute to the understanding of the complex relationships between geopolitical events and energy company stock prices?

1.5 Significance of the Study

This research holds significance for various stakeholders, including investors, financial analysts, policymakers, and industry experts. Understanding the dynamics of how geopolitical events, specifically the Russia-Ukraine war, reverberate through the stock markets of energy companies can provide valuable insights for risk assessment, portfolio management, and strategic decision-making [3]. Additionally, the findings may contribute to the development of more robust models for analyzing the impact of geopolitical risks on financial markets, with implications extending beyond the specific context of the Russia-Ukraine conflict [4].

1.6 Scope and Goals of the Research

Navigating the complex landscape of geopolitical events and their reverberations in financial markets requires a meticulous exploration of the Russia-Ukraine conflict's impact on global energy companies. This section delineates the overarching scope and specific goals that propel this research forward, shaping the foundation for a nuanced and comprehensive analysis.

1.6.1 Comprehensive Impact Assessment

At the heart of this study lies the ambition to conduct a thorough examination of how the Russia-Ukraine conflict has left an indelible imprint on the stock market performance of global energy companies. The scope extends beyond conventional analyses, delving into the intricate dynamics that unfold in the wake of geopolitical unrest.

1.6.2 Leveraging Principal Component Analysis (PCA)

A key goal of this research is to harness the analytical power of Principal Component Analysis (PCA). By doing so, we aim to unearth underlying patterns within complex financial datasets that conventional methodologies may overlook. PCA serves as a critical tool in deciphering the intricacies of the financial landscape, offering insights into the relationships and dynamics that emerge in times of crisis.

1.6.3 Identifying Crisis Points through Eigenvalues

An essential aspect of this study is the identification of crisis points through a meticulous examination of the first two eigenvalues. These eigenvalues serve as indicators of significant shifts in financial dynamics, offering a lens through which we can discern critical periods of change. Notably, the study places a particular emphasis on crisis points coinciding with major global events, including the pronounced changes observed during the COVID-19 pandemic following the Russia-Ukraine conflict in February 2022.

1.6.4 Heat Map Analysis for Crisis Dynamics

Building on the PCA findings, the research incorporates a heat map analysis, focusing on PCA1 changes. This analysis aims to unveil periods of heightened crisis, with specific attention to the sharp shifts observed during the COVID-19 pandemic post the Russia-Ukraine conflict. The heat map serves as a visual representation, providing a nuanced understanding of the intensity and duration of crisis dynamics.

1.6.5 Continent-Based Time Window Analysis

To further enrich our understanding, the study embarks on a continent-based time window analysis. Covering a period of six months from July 2018 to June 2023, this analysis involves assessing the Euclidean distance (R) of PC1 and PC2. The goal is to delineate how each continent faced heightened crisis instances, especially during the global upheaval caused by the COVID-19 pandemic that followed the Russia-Ukraine conflict.

1.6.6 Cluster Plot Visualization for Company Dynamics

Aiming to provide a tangible representation of the dynamic positions of companies before and after crisis points, this research utilizes cluster plots. These visualizations capture significant changes in company positions as they navigate the financial terrain during and after geopolitical and pandemic-induced crises. The cluster plot

analysis serves as a vivid portrayal of the evolving landscape for energy companies in the face of unprecedented challenges.

In summary, this section outlines the comprehensive scope and specific goals that drive the research forward. By combining analytical methodologies, including PCA, eigenvalue analysis, heat maps, and continent-based assessments, the study seeks to contribute a nuanced understanding of the intricate interplay between geopolitical events and global energy markets.

1.7 Structure of the Thesis

Following this introductory chapter, Chapter 2 provides an extensive literature review, integrating foundational concepts and previous studies that guide the trajectory of this research. Chapter 3 describes the theoretical background. Chapter 4 details the applied methodology, encompassing data collection, PCA application, and analytical approaches. Chapter 5 presents empirical findings, while Chapter 6 interprets and discusses these findings, drawing nuanced conclusions. The thesis concludes with Chapter 7, offering recommendations for future research avenues in this complex domain.

Chapter 2

Related Work

The relationship between geopolitical events and stock market performance has been the subject of extensive research in financial literature. Scholars have investigated how political tensions and conflicts can impact investor sentiment, market volatility, and asset prices. Notable studies by Khraiche (2023) [5] and Waszkiewicz et al. (2018) [6] emphasize the significance of considering geopolitical risk as a crucial factor in predicting stock market movements. These works provide a foundation for understanding the broader implications of political events on financial markets.

The nexus between armed conflicts and energy markets has garnered attention due to the inherent vulnerability of the energy sector to geopolitical disruptions. Research by Bouoiyour et al. (2019) [7] 3 underscores the susceptibility of energy prices to geopolitical shocks, particularly in regions critical to global energy production and distribution. Additionally, the work of Paravantis et al. (2019) [8] 4 delves into the spillover effects of geopolitical tensions on energy security, emphasizing the intricate linkages between global conflicts and energy markets.

To contextualize the impact of the Russia-Ukraine conflict on energy companies, this study adopts a theoretical framework rooted in financial and economic theories. The Efficient Market Hypothesis (EMH) posited by Fama (1970) [9] 5 serves as a foundational theory to understand how quickly and accurately financial markets incorporate new information, including geopolitical developments. Furthermore, Behavioral Finance theories, as explored by Shiller (2003) [10] 6, provide insights into the psychological factors influencing investor decisions during times of geopolitical uncertainty.

Principal Component Analysis (PCA) has gained prominence in financial research as a robust tool for analyzing complex datasets and identifying underlying patterns. In the study conducted by Waqar et al. (2017) [11] 7, PCA is employed to extract meaningful information from financial time series data. This methodology will be instrumental in uncovering the principal components driving the fluctuations in energy companies' stock prices during the Russia-Ukraine conflict.

Z. Zhang (Year) proposed a novel approach to stock price prediction using a PCA-LSTM model [12]. Zhang recommended employing PCA to reduce the dimensionality of fundamental data, integrating it with stock-related technical indicators such

as KDJ and MACD. The model, after modification in alignment with stock characteristics, demonstrated efficacy in generating predictions.
Y. Zhang et al. (Year) focused on forecasting crude oil futures market returns through a PCA combination approach [13]. Their research highlighted the utility of PCA in capturing essential patterns and information relevant to the crude oil market.
Waqar et al. (Year) contributed to predicting stock market trends using Principal Component Models (PCM) [11]. Their study emphasized that PCA can outperform linear regression when careful consideration is given to the relative correlation among input characteristics.

J. Choi et al. (Year) explored the asymptotic properties of correlation-based PCA, specifically addressing estimators of integrated eigenvalues and eigenvectors [14]. They provided analytical forms for the asymptotic biases and variances, enhancing the understanding of PCA applications in financial research.
A. Nobi & JW Lee (Year) investigated dynamic interactions and structural changes using PCA applied to cross-correlation coefficients of global financial indices from 1998 to 2012 [15]. They extended their analysis to explore financial states and group dynamics, employing PCA on cross-correlation coefficients of daily returns in commodity futures with S. Alam.
Omuya, E. O., Okeyo, G. O., & Kimwele, M. W. (Year) utilized PCA for feature selection and classification [16], showcasing the versatility of PCA beyond traditional financial market analyses.

Huang et al. (Year) introduced a scaled PCA approach for data dimension reduction, aiming to improve the traditional PCA by scaling predictors with their predictive slopes [17].
Studies related to the impact of external factors on the stock market have also been prevalent. Li, Y., Zhuang, X., Wang, J., & Dong, Z. (Year) investigated the effects of the COVID-19 pandemic on G20 stock markets [18], [19]. They employed Diebold and Yilmaz's (2012, 2014) DY approach and spatial econometric techniques to examine both direct and indirect impacts of the pandemic [20].
In the year 2022, Nerlinger M., & Utz S. studied the impact of the Russia-Ukraine conflict on energy firms from a capital market perspective, utilizing cumulative average abnormal returns (CAARs) [21]. CAARs, representing the sum total of abnormal returns, provided insights into the effects of the conflict on stock prices.

The literature review has equipped us with a thorough comprehension of pivotal concepts, theories, and methodologies pertinent to our study. Building upon this robust foundation, the forthcoming chapters will undertake empirical analysis, employing Principal Component Analysis (PCA) to evaluate the repercussions of the Russia-Ukraine conflict on global energy companies within stock markets. As we progress, the study aims to harness the power of PCA, utilizing eigenvalues, eigenvectors, and principal component coefficients to quantify the variance of both the stock market and crisis rates. This strategic approach advances the existing body of research, offering a distinctive contribution to unraveling the intricacies of financial markets during periods of crisis.

Chapter 3

Theoretical Background

3.1 Geopolitical Risks and Stock Markets

3.1.1 Theoretical Framework for Geopolitical Risks

Geopolitical risks have a profound impact on stock markets, influencing investor behavior, market volatility, and overall economic stability. Theoretical frameworks, such as the Political Economy theory and Behavioral Finance, help elucidate the mechanisms through which geopolitical events shape financial markets. The Political Economy theory posits that political events affect economic policies, thereby influencing market conditions. Additionally, Behavioral Finance explores the psychological factors driving investor decisions during times of geopolitical uncertainty.

3.1.2 Empirical Studies on Geopolitical Risks and Stock Markets

Building on theoretical foundations, empirical studies provide valuable insights into the real-world impact of geopolitical risks on stock markets. This section reviews seminal studies, examining methodologies and key findings. Notable research, such as that by A. Al-Maadid et al. [22], delves into the intricate relationship between geopolitical risk and international equity markets, offering empirical evidence to support theoretical frameworks.

3.2 Energy Sector Dynamics

3.2.1 Theoretical Foundations of Geopolitical Impact on Energy Markets

The energy sector, as a critical component of global economies, is particularly susceptible to geopolitical shocks. Theoretical frameworks in this section delve into how geopolitical events influence commodity prices, investment decisions, and the strategic behavior of energy companies [23]. Considerations include the Resource Curse theory, which explores the impact of geopolitical factors on resource-rich economies, and Game Theory, which provides insights into strategic interactions among energy-producing nations.

3.2.2 Energy Sector Resilience Strategies

In response to geopolitical risks, energy companies employ resilience strategies to mitigate potential disruptions. This subsection examines theoretical foundations related to diversification, hedging, and strategic resource management. Understanding how energy companies navigate geopolitical challenges is crucial for assessing their resilience in the face of events such as the Russia-Ukraine conflict.

3.3 Principal Component Analysis in Financial Analysis

3.3.1 Core Principles and Applications of PCA

Principal Component Analysis (PCA) is a powerful analytical tool in financial research for dimensionality reduction and pattern extraction [24]. This section provides an in-depth exploration of PCA's core principles, discussing how it is applied to financial time series data. Applications of PCA in portfolio optimization, risk management, and anomaly detection in financial markets are highlighted, demonstrating the versatility of this method.

3.3.2 Critiques and Limitations of PCA

While PCA is widely used, it is essential to acknowledge its critiques and limitations. This subsection discusses potential challenges, such as sensitivity to outliers and assumptions underlying PCA. A critical understanding of these limitations is crucial for interpreting results accurately.

3.4 Conceptual Model

Building on the theoretical foundations of geopolitical risks, energy sector dynamics, and PCA, this section presents a conceptual model that frames the study's approach. The model outlines the interconnectedness of geopolitical indicators, such as conflict intensity, economic sanctions, and energy supply disruptions, with the stock market performances of energy companies. Additionally, it highlights the role of PCA in capturing the multidimensional nature of these relationships, offering a structured framework for empirical analysis. The conceptual model guides the research in developing hypotheses and selecting relevant variables for investigation.

In summary, this chapter establishes the theoretical framework for the study by exploring the links between geopolitical risks and stock markets, elucidating the dynamics of the energy sector, introducing PCA as a powerful analytical tool, and presenting a conceptual model that integrates these elements. The theoretical foundation provides the groundwork for the empirical investigation into the impact of the Russia-Ukraine war on global energy company stocks using PCA in subsequent chapters.

Chapter 4

Methodology

4.1 Data Collection

The dataset utilized in this thesis spans from July 2018 to June 2023, capturing a critical period marked by significant global events, including the COVID-19 pandemic and the Russia-Ukraine war. To ensure a comprehensive representation, we deliberately selected 50 energy companies from different continents, specifically focusing on top-category companies within their respective regions or countries. The data, meticulously sourced from Yahoo Finance [25], with a primary focus on daily stock market attributes. The dataset details the daily opening and closing prices of the selected energy companies, providing a comprehensive overview of their financial performance during a period of notable geopolitical and economic challenges.

The inclusion of data during the COVID-19 crisis, which unfolded in early 2020, adds a layer of complexity to the analysis. The pandemic had widespread implications for global markets, affecting various industries, including the energy sector. The dataset covers the phases of market volatility, disruptions in supply chains, and economic uncertainties associated with the pandemic.

Furthermore, the dataset encompasses the period of the Russia-Ukraine war, which emerged in February 2022. This geopolitical crisis introduced heightened uncertainties in the global arena, particularly impacting the energy sector due to its sensitivity to geopolitical dynamics. By incorporating data during this period, the analysis aims to provide insights into how energy companies navigated and responded to the challenges posed by the Russia-Ukraine conflict.

The dataset was meticulously constructed through web scraping techniques, ensuring the inclusion of valuable information for analyzing the dynamics of the energy sector, studying stock performance, and investigating various financial metrics relevant to energy companies. This comprehensive dataset serves as the foundation for the empirical analysis, allowing for a nuanced examination of the interplay between geopolitical events, such as the COVID-19 pandemic and the Russia-Ukraine war, and the stock market performances of energy companies.

Table 4.1 presents a comprehensive dataset encompassing information for 50 energy companies. The dataset includes the following key columns

- Company Name: This column lists the names of the selected energy companies. These companies are chosen based on their significance in the global energy sector.
- Ticker: The Ticker column provides the stock ticker symbols associated with each energy company. Ticker symbols are unique abbreviations used in stock exchanges to identify a specific publicly traded company.
- Country: This column indicates the country where each energy company is headquartered or primarily operates. The geographic location is a crucial factor in understanding the regional dynamics that may influence a company's operations.
- Continent: The Continent column specifies the continent to which the country of each energy company belongs. This information is essential for analyzing global trends and understanding regional variations in the energy sector.
- Key Name: The Key Name column is a concatenated key that combines the short form of the continent, the short form of the country, and the ticker symbol. This unique identifier is designed to facilitate efficient data sorting and retrieval. For instance, the key name "NA_U_XYZ" represents a company with the ticker symbol "XYZ" operating in the United States (U) within the North American continent (NA).

Table 4.1: Dataset Table

Company Name	**Ticker**	**Country**	**Continent**	**Key Name**
Oando PLC	OAO.JO	Nigeria	Africa	AF_N_OAO.JO
Sasol Limited	SSL	South Africa	Africa	AF_S_SSL
CNOOC Limited	0883.HK	China	Asia	A_C_0883.HK
PetroChina Company Limited	601857.SS	China	Asia	A_C_601857.SS
Indian Oil Corporation Limited	IOC.NS	India	Asia	A_I_IOC.NS
Reliance Industries Limited	RELIANCE.NS	India	Asia	A_I_RELIANCE.NS
ENEOS Holdings, Inc	5020.T	Japan	Asia	A_J_5020.T
Japan Petroleum Exploration Co., Ltd.	1662.T	Japan	Asia	A_J_1662.T
Formosa Petrochemical Corporation	6505.TW	Taiwan	Asia	A_T_6505.TW
PTT Public Company Limited	PTT.BK	Thailand	Asia	A_TH_PTT.BK
PETRONAS Gas Berhad	6033.KL	Malaysia	Asia	A_M_6033.KL
S-Oil Corporation	010950.KS	South Korea	Asia	A_S_010950.KS

Independent Petroleum Group K.S.C.P.	IPG.KW	Kuwait	Asia	A_K_IPG.KW
AGL Energy Limited	AGL.AX	Australia	Australia	AU_A_AGL.AX
Origin Energy Limited	ORG.AX	Australia	Australia	AU_A_ORG.AX
Santos Limited	STO.AX	Australia	Australia	AU_A_STO.AX
OMV Aktiengesellschaft	OMV.VI	Austria	Europe	E_A_OMV.VI
Ørsted A/S	ORSTED.CO	Denmark	Europe	E_D_ORSTED.CO
Neste Oyj	NESTE.HE	Finland	Europe	E_F_NESTE.HE
TotalEnergies SE	TTE	France	Europe	E_FR_TTE
E.ON SE	EOAN.DE	Germany	Europe	E_G_EOAN.DE
MOL Hungarian Oil and Gas PLC	MOGB.BE	Hungary	Europe	E_H_MOGB.BE
Eni S.p.A.	ENI.MI	Italy	Europe	E_I_ENI.MI
Equinor ASA	EQNR	Norway	Europe	E_N_EQNR
Orlen S.A.	PKN.WA	Poland	Europe	E_P_PKN.WA
Repsol, S.A.	REP.MC	Spain	Europe	E_S_REP.MC
BP p.l.c.	BP	United Kingdom	Europe	E_U_BP
Capricorn Energy PLC	CNE.L	United Kingdom	Europe	E_U_CNE.L
Tullow Oil plc	TLW.L	United Kingdom	Europe	E_U_TLW.L
Galp Energia, SGPS, S.A.	GALP.LS	Portugal	Europe	E_P_GALP.LS
Canadian Natural Resources Ltd	CNQ	Canada	North America	NA_C_CNQ
Enbridge Inc.	ENB	Canada	North America	NA_C_ENB
Suncor Energy Inc.	SU	Canada	North America	NA_C_SU
TC Energy Corporation	TRP	Canada	North America	NA_C_TRP
Chevron Corporation	CVX	United States	North America	NA_U_CVX
ConocoPhillips	COP	United States	North America	NA_U_COP
Energy Transfer LP	ET	United States	North America	NA_U_ET
Enterprise Products Partners L.P.	EPD	United States	North America	NA_U_EPD
EOG Resources, Inc.	EOG	United States	North America	NA_U_EOG

Exxon Mobil Corporation	XOM	United States	North America	NA_U_XOM
Kinder Morgan, Inc.	KMI	United States	North America	NA_U_KMI
Occidental Petroleum Corporation	OXY	United States	North America	NA_U_OXY
Phillips 66	PSX	United States	North America	NA_U_PSX
Valero Energy Corporation	VLO	United States	North America	NA_U_VLO
Williams Companies, Inc.	WMB	United States	North America	NA_U_WMB
Pampa Energía S.A.	PAM	Argentina	South America	SA_A_PAM
Sociedad Anónima	YPF	Argentina	South America	SA_A_YPF
Petroleo Brasileiro S.A. (Petrobras)	PBR	Brazil	South America	SA_B_PBR
Ultrapar Participações S.A.	UGP	Brazil	South America	SA_B_UGP
Ecopetrol S.A.	EC	Colombia	South America	SA_C_EC

This dataset, with its diverse set of companies and key attributes, serves as a foundation for the empirical analysis in this study. The inclusion of geographical and key identification information enables a nuanced exploration of how energy companies from different regions respond to geopolitical events and market dynamics.

4.2 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a powerful statistical technique widely employed in finance and data analysis to reveal the underlying structures and patterns within multivariate datasets. In the context of examining the daily logarithmic returns of energy companies' stock prices, PCA serves as a robust tool to distill complex market dynamics into a set of uncorrelated variables known as principal components. This analytical approach enables the identification of key factors influencing market behavior, providing valuable insights into the relative contributions of individual energy companies and their impact on overall market conditions.

Steps in the Principal Component Analysis (PCA) Procedure:

4.2.1 Definition of Daily Logarithmic Return

Begin by calculating the daily logarithmic return ($r_i(t)$) for each energy company i. This step captures the percentage change in stock prices by taking the natural logarithm of the ratio of closing prices between consecutive days.

The formula for $r_i(t)$ is given by:

$$\begin{aligned} r_i(t) &= \ln[P_i(t)] - \ln[P_i(t-1)] \\ \text{where } P_i(t) &\text{ is the closing price of energy company } i \text{ on day } t. \end{aligned} \tag{4.1}$$

4.2.2 Normalization of Returns

Normalize the daily returns ($r\acute{}(t)$) by subtracting the mean ($\langle \mathrm{r_i} \rangle$) and dividing by the standard deviation (σ_i). This normalization ensures that returns are on a comparable scale, facilitating meaningful comparisons.
The normalized return is defined as:

$$r_i'(t) = \frac{r_i(t) - \langle r_i \rangle}{\sigma_i} \tag{4.2}$$

where σ_i is the standard deviation of the time series r_i given by:

$$\sigma_i = \sqrt{\langle r_i^2 \rangle - \langle r_i \rangle^2} \tag{4.3}$$

4.2.3 Construction of the Return Matrix

Organize the normalized returns into a matrix R with dimensions NT, where N represents the total number of energy companies, and T represents the total number of days. This matrix forms the foundation for subsequent analysis. The standardized matrix R is given by:

$$R = \begin{bmatrix} r_1'(t_1) & \dots & r_1'(T) \\ \vdots & \ddots & \vdots \\ r_N'(t_1) & \dots & r_N'(T) \end{bmatrix} \tag{4.4}$$

where N is the total number of energy companies and T is the total number of days in a time window. The symbol $\langle \cdots \rangle$ denotes the average over the time window.

4.2.4 Calculation of Empirical Correlation Matrix

Compute the empirical correlation matrix (C) by multiplying the normalized return matrix (R) by its transpose (R^T), and then dividing by the total number of days (T). This matrix captures the relationships and dependencies between different energy companies.
It is used to calculate the empirical correlation matrix as follows:

$$C = \frac{1}{T} R R^T \tag{4.5}$$

where R^T is the transpose of the matrix R and elements of the correlation matrix $C_{ij} \in [-1, 1]$. Because we have standardized the time series, the empirical correlation matrix C of the return matrix R is equal to the covariance matrix of R.

4.2.5 Eigenvalue Decomposition of Correlation Matrix

Perform eigenvalue decomposition on the correlation matrix (C) to obtain three matrices: V,M, and V^T. M is a diagonal matrix containing eigenvalues (λ_i) in descending order, and V is an orthogonal matrix containing corresponding eigenvectors.
Diagnose the $N \times N$ correlation matrix C in the form:

$$C = VMV^T \quad (5) \tag{4.6}$$

where M is the diagonal matrix with the diagonal elements as eigenvalues $\lambda_i = (\lambda_1, \ldots, \lambda_N)$ in descending order, and V is an orthogonal matrix of the corresponding eigenvectors.

4.2.6 Covariance Matrix of Returns with PCs

The covariance matrix $\mathrm{Cov}(Y, R)$ of the returns matrix R with PCs Y is expressed in terms of the orthogonal matrix V, the diagonal matrix M containing eigenvalues, and their transposes. This relationship is crucial for understanding how the PCs capture the variance in the original return matrix [26]. The covariance matrix of the return R with PCs Y can be written as:

$$\mathrm{Cov}(Y, R) = \frac{1}{T} YRR^T = \frac{1}{T} VRR^T = VC = VVV^T MV = MV \tag{4.7}$$

Again, each eigenvalue and the corresponding eigenvector can be written as:

$$\lambda_i = v_i^T C v_i = v_i \mathrm{Cov}(R_t) v_i^T = \mathrm{Var}(v_i^T R_t) = \mathrm{Var}(y_{k,t}) \tag{4.8}$$

where $y_k = v_k^T R$ is known as the k-th principal component [27]. The eigenvalue λ_i indicates the portion of the total variance of R contributed to the principal component y_k.

4.2.7 Cross Correlation of Returns and PCs

The cross-correlation $\mathrm{Corr}(r_i', y_{1,t})$ between the returns of asset i and the k-th principal component y_k is represented as $\frac{1}{\lambda_i} v_{i1}$, where λ_i is the eigenvalue and v_{i1} is the PC coefficient. These correlations are scaled by the appropriate eigenvalue, providing a measure of the contribution of each index to the principal components.

4.2.8 Temporal Change in Correlations and PC Coefficients

The temporal change in correlations or PC coefficients is crucial for identifying market states. Examining the correlations between indices and PCs for the largest eigenvalues allows for a determination of the prevailing market conditions. The significant PCs, often associated with the largest eigenvalues, are particularly informative about market dynamics.

To quantify the change in market conditions, the difference in correlations for each index contributing to PC1 between two time periods is defined as

$$\Delta C_1(t_1, t_2) = C_1(t_2) - C_1(t_1). \tag{4.9}$$

This measure enables the comparison of market states at different times, with high positive or negative values indicating distinct market conditions.

Interpreting ΔC_1 Values

- High negative values of ΔC_1 suggest that two markets at different times are significantly different, indicating a shift towards a calmer state.
- High positive values of ΔC_1 imply a substantial difference between markets, signaling an entry into a crisis state.
- Smaller ΔC_1 values indicate similarity in market states between the two time periods.

In essence, the analysis of ΔC_1 provides a dynamic tool for tracking and comparing market conditions over time, offering valuable insights into shifts between crisis and non-crisis states. It allows for the identification of temporal changes in correlations, aiding in the characterization of market states based on the behavior of the largest principal components.

Chapter 5

Result

The Results chapter unfolds a nuanced exploration into the aftermath of the Russia-Ukraine war on global energy companies, employing a multifaceted analytical approach. From the revelation of key eigenvalues to the identification of crisis states, the visualization of temporal dynamics, and the cluster-based categorization of companies, this chapter delves deep into the intricate tapestry of market responses and adaptive strategies within the energy sector.

5.1 Eigenvalue Exploration: Decoding Dominant Market Factors

In the quest to decipher the underlying dynamics of the market for energy companies, we turn our attention to the eigenvalues and eigenvectors of the correlation matrices. These eigenvalues play a pivotal role in unraveling the intricate interplay of market forces, with a particular focus on the largest and second-largest eigenvalues, which encapsulate valuable information on group dynamics.
Figure 5.1 provides a compelling visual narrative of the time evolution of the first and second eigenvalues within half-year correlation matrices. The first eigenvalue, being more informative due to its fewer abrupt transitions, becomes a key focal point for our analysis.

July 2018 to December 2019: Stability Amidst Minimal Variations:

The initial phase, spanning from July 2018 to December 2019, depicts a market characterized by minimal variations in eigenvalues. This period suggests a stable condition in the stock market for energy companies over six-month intervals.

January–June 2020: Transition and the Onset of COVID-19 Impact:

Noteworthy transitions unfold during January–June 2020, marking a pivotal moment. The first eigenvalue, after a slight decrease in the latter half of 2019, experiences a sharp increase in the first half of 2020. This surge aligns with the onset of the COVID-19 crisis, indicating the profound impact of the pandemic on market conditions.

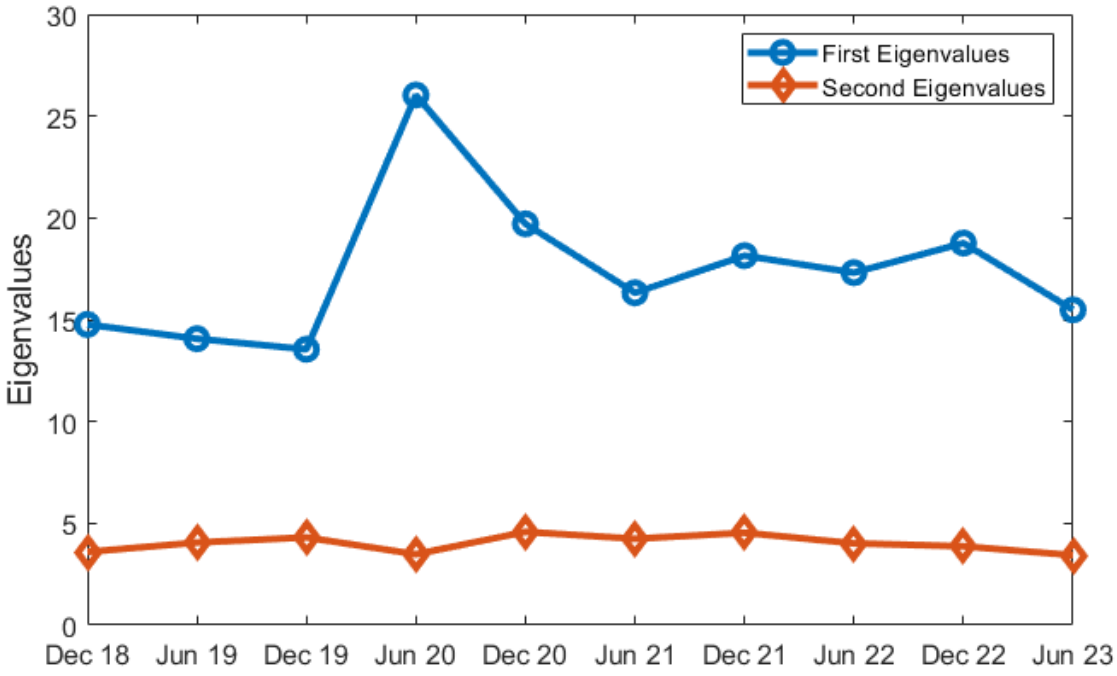


Figure 5.1: Time Evolution of Eigenvalues

July–December 2020: COVID-19 Pandemic's Ongoing Impact:

The year 2020 unfolds as a transformative period, particularly from January to December. The first half witnessed a notable increase in the first eigenvalue, while the second half experienced a significant decrease. This fluctuation reflects the turbulent market conditions brought about by the persisting impact of the COVID-19 pandemic.

July 2021–December 2022: Fluctuations Amidst Ongoing Changes:

From July 2021 to December 2022, the eigenvalues exhibit slight fluctuations marked by both increases and decreases. This period signifies ongoing changes in market conditions, potentially influenced by a combination of external factors. It's crucial to note that these fluctuations coincide with the Russia-Ukraine war, adding a layer of geopolitical complexity to the market dynamics during this specific timeframe. The interplay of economic and geopolitical forces during this period contributes to the nuanced fluctuations observed in the eigenvalues, shaping the trajectory of market changes in response to the evolving global landscape.

January–June 2023: A Glimpse into the Later Period:

In the latter half of January–June 2023, a slight decrease in eigenvalues is observed, adding a nuanced layer to the evolving narrative. This observation hints at potential shifts in market dynamics during this specific time frame.

In essence, the time evolution of eigenvalues becomes a compelling lens through which we observe the transformative journey of the market for energy companies,

capturing stability, abrupt transitions, and the persistent impact of external events, particularly the COVID-19 pandemic. This exploration sets the stage for a more in-depth analysis of the ensuing results, unraveling the intricate responses and adaptations of energy companies within this dynamic landscape.

5.2 Crisis-State Identification Heatmap: Temporal Evolution of Challenges

In our endeavor to unravel the temporal evolution of challenges faced by energy companies, the analysis pivots toward a Crisis-State Identification Heatmap. This heatmap becomes an intricate canvas, painted with the normalized returns of commodities and principal components (PCs), offering a nuanced perspective on the unfolding market states.

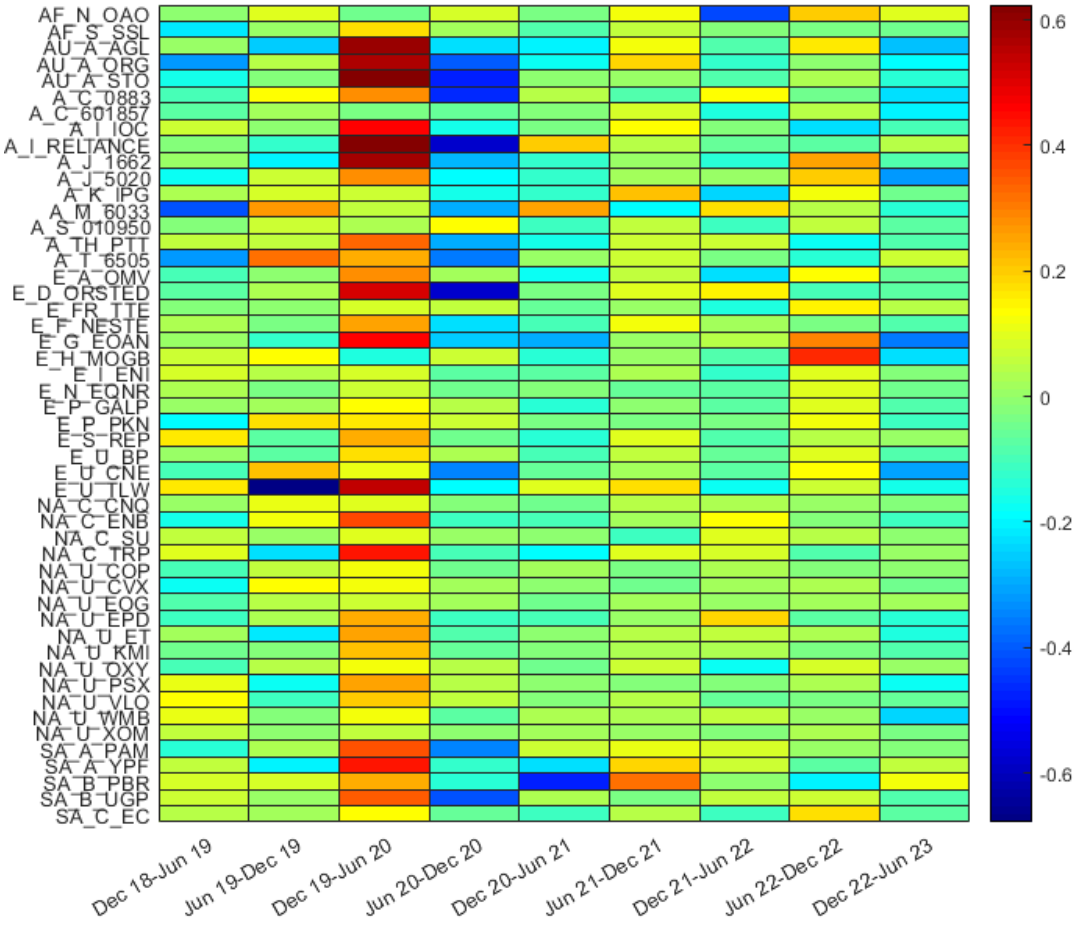


Figure 5.2: Dynamics of PC Coefficients Heatmap

Figure 5.2 encapsulates the dynamic evolution of principal component (PC) coefficients, offering a granular portrayal of changes in PC1 values across successive time windows. This heatmap serves as a visual chronicle, illuminating the nuanced shifts in market dynamics as reflected in the alterations of PC1 values from one time window to the next.

Temporal Variation in PC1 Values:

The color-coded heatmap visually articulates the temporal dynamics of PC1 coefficients, capturing the alterations in values over distinct 6-month intervals. Each cell in the heatmap represents the PC1 coefficient for a specific commodity during a particular time window, allowing for a detailed examination of how these coefficients fluctuate over time.

Latter Half of 2019 and Initial Half of 2020 (COVID-19 Impact):

In the period spanning the latter half of 2019 and the initial half of 2020, the Crisis-State Identification Heatmap reveals notable shifts in PC1 coefficients, indicative of the profound impact of the COVID-19 crisis on energy companies. This temporal segment is characterized by a discernible transition, visually depicted in the heatmap's color-coded representation of PC1 values. The intensity and frequency of changes in PC1 values during this turbulent period underscore the heightened market sensitivity, emphasizing the dynamic and responsive nature of energy markets in the face of external challenges.
Ten companies, representing diverse continents, were identified as facing significant crises during this timeframe, list show in Table 5.1. These companies, spanning Asia, Australia, Europe, and North America, provide a global perspective on the challenges experienced by energy corporations in the face of the COVID-19 pandemic.

Table 5.1: Selected Companies Facing Significant Crises during Latter Half of 2019 and Initial Half of 2020

Company Name	**Ticker**	**Country**	**Continent**	**Final Name**
Reliance Industries Limited	RELIANCE.NS	India	Asia	A_I_RELIANCE
Santos Limited	STO.AX	Australia	Australia	AU_A_STO
AGL Energy Limited	AGL.AX	Australia	Australia	AU_A_AGL
Japan Petroleum Exploration Co., Ltd.	1662.T	Japan	Asia	A_J_1662
Origin Energy Limited	ORG.AX	Australia	Australia	AU_A_ORG
Tullow Oil plc	TLW.L	United Kingdom	Europe	E_U_TLW
Ørsted A/S	ORSTED.CO	Denmark	Europe	E_D_ORSTED
E.ON SE	EOAN.DE	Germany	Europe	E_G_EOAN
Indian Oil Corporation Limited	IOC.NS	India	Asia	A_I_IOC
TC Energy Corporation	TRP	Canada	North America	NA_C_TRP

This diverse group of companies faced challenges reflective of the global economic turmoil caused by the COVID-19 pandemic. The Table 5.1 provides a succinct overview of their names, stock tickers, home countries, and continents, emphasizing the varied geographic impact of the crisis on these entities.

First and Last Halves of 2020 (Recovery Phase):

After the crisis, both the first and last halves of 2020 reveal an upward trend in PC1 coefficients. This indicates a phase of recovery and resilience among energy companies, emphasizing their ability to adapt to changing market conditions.

Latter Half of 2021 to Initial Half of 2022 (Stability):

The heatmap illustrates a period of relative stability in PC1 coefficients during the latter half of 2021 to the initial half of 2022. The subdued variations in color suggest a more consistent market environment, where PC1 values exhibit less pronounced fluctuations.

First Half to Latter Half of 2022 (Post-Russia-Ukraine War):

The transition from the first half of 2022 to the latter half, corresponding to the aftermath of the Russia-Ukraine war, revealed a notable change in PC1 values. This shift underscored how companies responded to geopolitical events, showcasing their adaptability in navigating through post-crisis scenarios. During this period, three companies experienced the maximum crisis, exhibiting significant changes in their PC1 values (see Table 5.2).

Table 5.2: Companies Experiencing Maximum Crisis Post-Russia-Ukraine War (First Half to Latter Half of 2022)

Company Name	**Ticker**	**Country**	**Continent**	**Final Name**
MOL Hungarian Oil and Gas PLC	MOGB.BE	Hungary	Europe	E_H_MOGB
E.ON SE	EOAN.DE	Germany	Europe	E_G_EOAN
Japan Petroleum Exploration Co., Ltd.	1662.T	Japan	Asia	A_J_1662

These companies, spanning Europe, Asia represent instances where the aftermath of the Russia-Ukraine war had a pronounced impact, leading to heightened crisis conditions. The selected companies serve as notable examples of entities facing significant challenges during the specified period, providing insights into the varied responses of companies from different continents in the post-crisis landscape.

In essence, Figure 5.2 unfolds a detailed narrative of market dynamics by examining the temporal evolution of challenges faced by energy companies through the lens of PC1 coefficients. The color gradients within the heatmap provide a visual context, offering valuable insights into the evolving nature of market conditions. This portrayal highlights the responsiveness and adaptability of energy companies across distinct time intervals. The fluctuations in PC1 values, vividly depicted in the color-coded heatmap, serve as a powerful tool for understanding how market dynamics change over time, providing a comprehensive view of the challenges and adaptive strategies adopted by energy corporations.

5.3 Cluster-Based Categorization: Understanding Adaptive Strategies

In this pivotal chapter, we embark on a journey to unravel the adaptive strategies employed by 50 global energy companies in the aftermath of the Russia-Ukraine war. At the core of our exploration is a sophisticated cluster-based analysis, leveraging a cluster plot based on PC1 and PC2 values. This plot, crafted with the meticulous use of the Ward method for clustering, serves as our lens into the intricate responses of energy corporations. Notably, PC1 and PC2 encapsulate rich dynamic data, with PC1 being particularly informative.

Employing this cluster plot, the x-axis unfolds the variance along the PC1 value, while the y-axis articulates the nuances within the PC2 value. Each data point on the plot corresponds to an individual energy company, and their continental affiliations are meticulously denoted by distinct color codes (A=Asia, AU=Australia, E=Europe, AF=Africa, NA=North America, and SA=South America). We delve into the cluster-based categorization during two distinct crisis states, namely the Covid-19 pandemic and the Russia-Ukraine war.

5.3.1 Cluster Dynamics Preceding and During the Covid-19 Pandemic

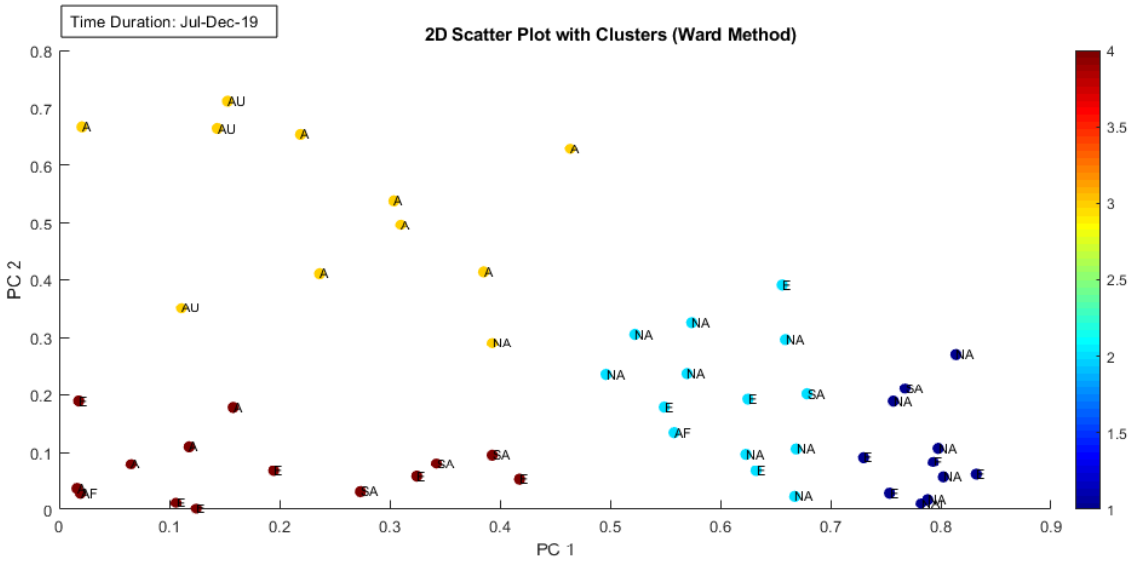


Figure 5.3: Plot Graph for the Last Half of 2019 (Pre-Covid Stage)

In Figure 5.3, the plot graph illustrates the distribution of all companies in the last half of 2019, preluding the onset of the Covid-19 pandemic. At this time, all companies are dispersed across four distinct clusters. Each cluster harbors companies from various continents, and the color-coded points represent their geographical affiliations. Notably, the yellow cluster predominantly comprises companies from Asia and Australia, while the light blue cluster encompasses North American companies,

with a mix of European and South American entities. The deep blue cluster involves North American, South American, and European companies, while the deep red cluster comprises companies from Europe, South America, and Asia.

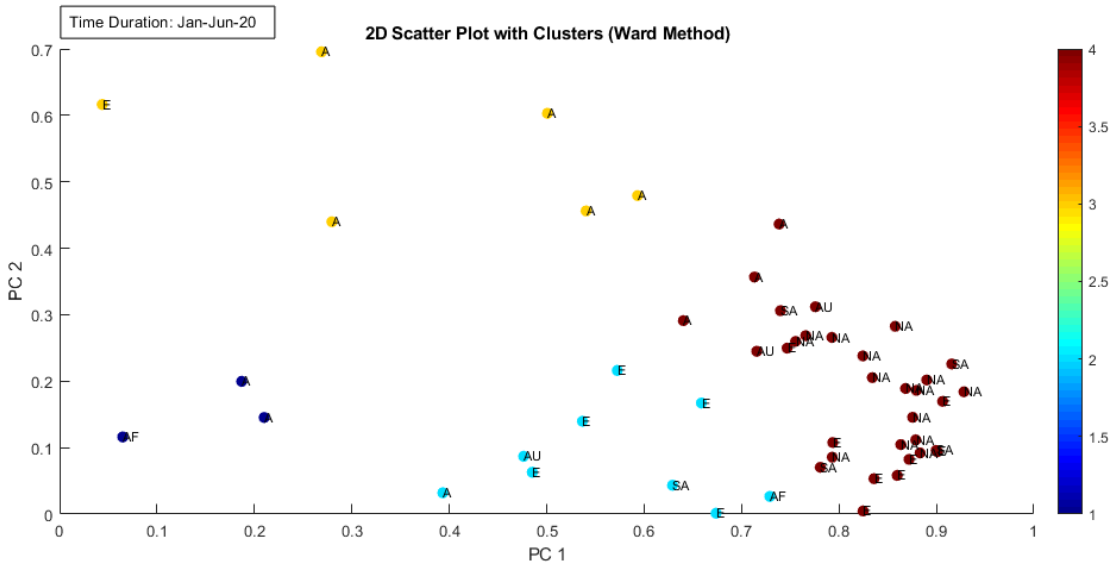


Figure 5.4: Plot Graph for the First Half of 2020 (COVID-19 Impact)

Figure 5.4 portrays the plot graph during the first half of 2020, encapsulating the impact of the Covid-19 pandemic. Similar to the pre-crisis period, four clusters emerge based on PC1 and PC2 values. The yellow cluster is dominated by Asian regions' companies, the deep blue cluster involves only three companies (two Asian and one African), the light blue cluster contains predominantly European companies, and the deep red cluster encompasses companies from all continents.

Analyzing both Figures 5.3 and 5.4, a noticeable shift is evident. Before the crisis, companies were dispersed, while during the crisis, they coalesced into clusters. Companies that were initially closer to the origin point with lower PC1 values tend to shift further away during the crisis, signifying a transition. Notably, almost every continent, excluding North America, experiences a crisis during this period.

5.3.2 Cluster Dynamics Preceding and During the Russia-Ukraine War

Figure 5.5 presents the plot graph for the first half of 2022, preceding the Russia-Ukraine war. Similar to the pre-Covid stage, companies are distributed across four clusters. The deep red cluster is dominated by companies from Asia and Australia, the yellow cluster involves companies from Asia, Europe, and other continents, the light blue cluster encompasses North American, South American, and European companies, and the deep blue cluster comprises mainly North American companies.

Figure 5.6 depicts the plot graph during the second half of 2022, reflecting the impact of the Russia-Ukraine war. Four clusters persist, with the yellow cluster being

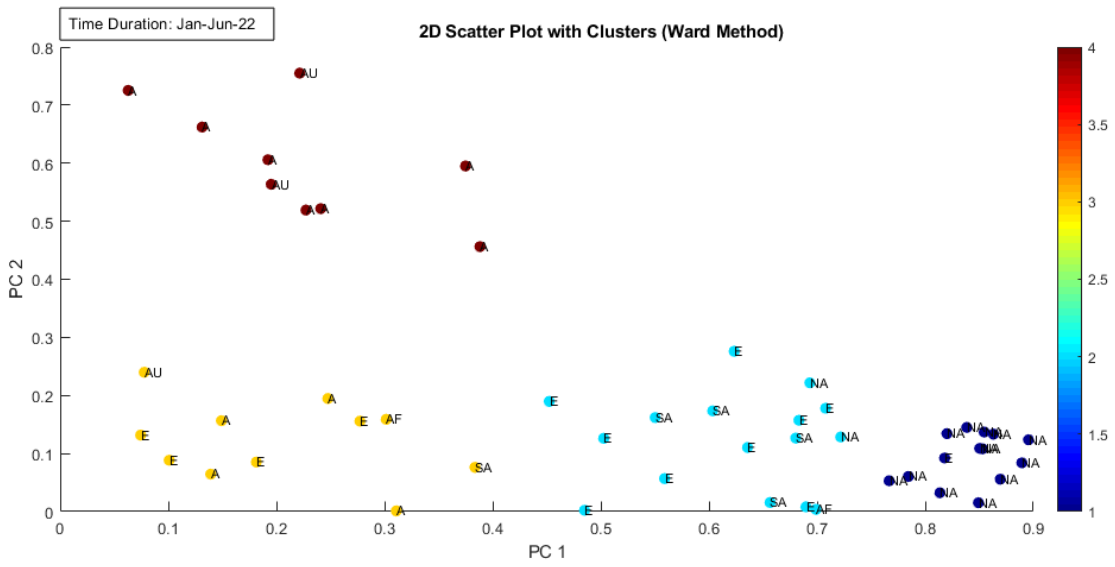


Figure 5.5: Plot Graph for the First Half of 2022 (Pre-Russia-Ukraine War)

dominated by Asian regions' companies, the deep red cluster involving Asian, Australian, and one African company, the light blue cluster containing predominantly European companies, and the deep blue cluster comprising more North American companies but also featuring some European entities.

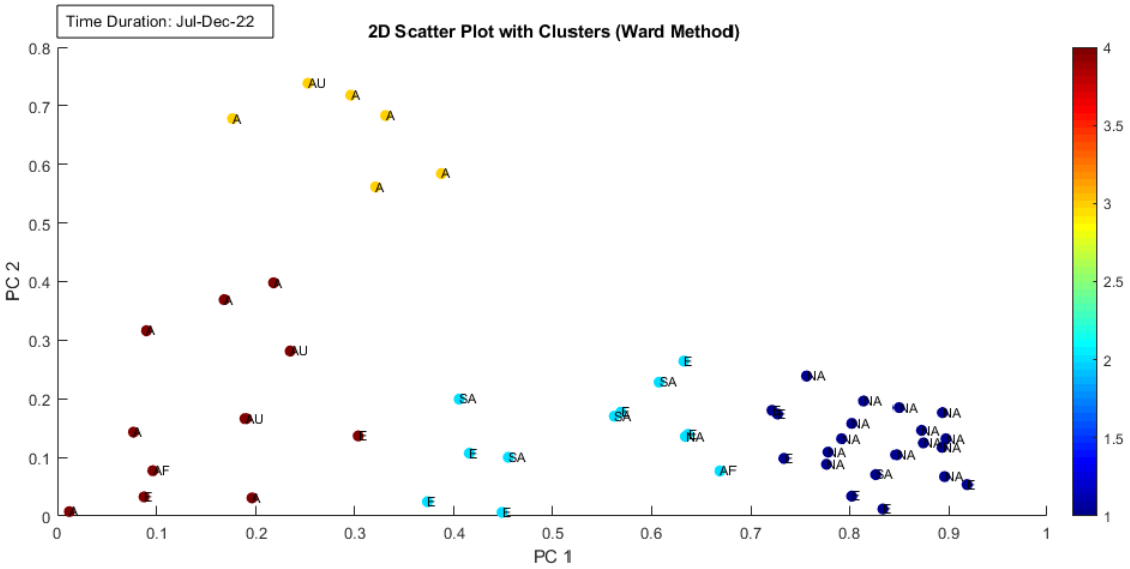


Figure 5.6: Plot Graph for the Second Half of 2022 (Russia-Ukraine War Impact)

Comparing Figures 5.5 and 5.6, minimal changes in clusters are observed, indicating slight alterations. Only a few companies experience shifts in their PC1 values, signifying their entry into a crisis state.

This in-depth analysis through cluster-based categorization unveils the nuanced adaptive strategies adopted by energy companies during the COVID-19 pandemic and the Russia-Ukraine war. The spatial transitions and clustering patterns offer

valuable insights into how companies strategically positioned themselves in response to geopolitical uncertainties.

5.4 Dynamics Visualization: Continental-Based Responses in the Principal Component Space

As the narrative unfolds into a graphical representation through the R graph, we delve into a nuanced exploration of temporal dynamics measured by the Euclidean distances derived from PC1 and PC2. This distinctive visualization offers a dynamic lens, plotting the average continent-based Euclidean distances in the principal component space. Through this representation, the graph becomes a holistic view of how continents respond to evolving crises, capturing the interconnectedness of global markets over time.

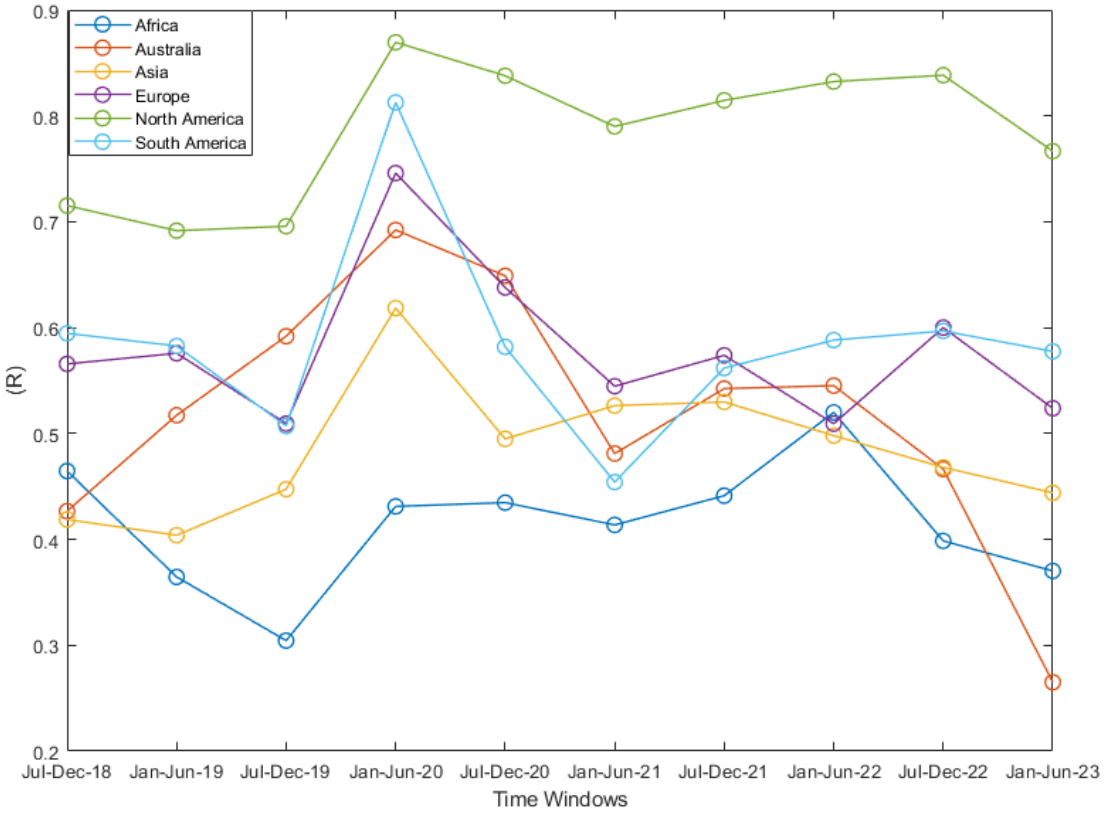


Figure 5.7: Dynamics of Crisis States: R Evolution in PC1-PC2 Space

In the exploration of continental responses within the principal component space, Figure 5.7 takes center stage, presenting the graph of R versus time window. The Euclidean distance of PC1 and PC2, denoted as 'R' becomes a crucial metric in understanding the evolving dynamics over distinct time intervals. In regions where there is a pronounced shift in the value of R, such changes signify potential crisis states. An increase from a lower value implies a crisis state, while a decrease indicates a return to normalcy.

Key Insights from Figure 5.7:

COVID-19 Crisis:

The graph prominently illustrates the average R-value for the Sacom continent, showcasing a sharp change during the period of the COVID-19 crisis. The noticeable increase in R signifies a crisis state, underlining the substantial impact on the market dynamics of Sacom. The subsequent normalization suggests a gradual recovery.

Russia-Ukraine War:

Comparatively, the graph indicates a milder change during the Russia-Ukraine war, signifying a period of crisis but to a lesser extent. The fluctuations in R values across all continents during this period suggest a heightened state of uncertainty, with values reflecting a shift from the normal state.

European Crisis during Russia-Ukraine War:

Notably, Europe's R-value experiences a more significant increase during the Russia-Ukraine war, indicating a heightened crisis for European companies. This specific regional insight sheds light on the differentiated impact of geopolitical events on various continental markets.

Post-Crisis Normalization:

The gradual decrease in R values post both the COVID-19 crisis and the Russia-Ukraine war indicates a return to normalcy, albeit with varying rates of normalization across continents. This observation highlights the resilience of global markets in recovering from external shocks.

Figure 5.7 serves as a dynamic visual narrative, capturing the temporal evolution of crisis states and recovery phases across continents. The sharp changes in R values provide insights into the intensity and duration of crises, offering a comprehensive understanding of how continental markets respond to external shocks.

In essence, this graphical representation becomes a powerful tool for discerning continental-based responses in the principal component space, shedding light on the nuanced impact of crises on global markets and providing valuable insights for strategic decision-making in the face of economic uncertainties.

Chapter 6

Discussion

This chapter delves into a comprehensive discussion of the findings extracted from our in-depth analysis of the repercussions of the Russia-Ukraine war on global energy companies. Our investigation spans eigenvalue exploration, crisis-state identification, dynamics visualization, and cluster-based categorization. Within the discourse, we uncover the limitations of our study and propose potential avenues for future research.

In scrutinizing the key findings, the time evolution of eigenvalues emerges as a pivotal focus, showcasing distinct transitions during the COVID-19 pandemic and the Russia-Ukraine war. The Crisis-State Identification Heatmap provides a nuanced perspective on the temporal evolution of challenges faced by energy companies, elucidating significant shifts during crises and subsequent recovery phases. The graphical representation of continental responses in the principal component space offers insights into the interconnectedness of global markets, emphasizing the post-crisis normalization and resilience of the energy sector.

However, acknowledging the study's constraints is crucial. Data limitations and the multifaceted nature of geopolitical dynamics serve as constraints that impact the precision and comprehensiveness of our analysis. These limitations underscore the need for cautious interpretation of our findings.

Looking forward, our study lays the groundwork for future research endeavors. Expanding the dataset to encompass a broader spectrum of energy companies and incorporating more geopolitical events will enhance the depth and breadth of our analysis. Leveraging advanced analytical techniques, such as machine learning algorithms, can unearth hidden patterns, providing a more nuanced understanding of market responses. Additionally, delving into the policy implications of adaptive strategies during crises can offer valuable insights into the resilience of the energy sector and inform strategic policy decisions.

In essence, this discussion chapter synthesizes the intricate findings of our analysis, acknowledges study limitations, and charts a course for future research. The nuanced exploration of energy markets post-crisis sets the stage for continued inquiry into the adaptive strategies of global energy companies within the ever-evolving geopolitical landscape.

Chapter 7

Conclusion

In the pursuit of understanding the effects of the Russia-Ukraine conflict and the COVID-19 pandemic on global energy companies, this research took an unexpected turn, revealing a predominant influence of the latter. Despite our initial focus on analyzing the Russia-Ukraine impact, the findings underscore the unparalleled disruptions caused by the COVID-19 pandemic in the energy sector. The temporal exploration of eigenvalues, crisis-state identification heatmap, and continental-based responses highlighted the dynamic and evolving nature of market conditions—notably, fluctuations observed during the Russia-Ukraine conflict aligned with broader economic and geopolitical shifts.

Embarking on a journey from the exploration of eigenvalues to the identification of crisis states and concluding with a cluster-based categorization, the narrative distinctly pivoted towards the seismic disruptions induced by the COVID-19 pandemic. The eigenvalues initially portrayed a semblance of stability in the market, only to witness an abrupt transition with the onset of the pandemic, setting the stage for a tumultuous period. In stark contrast, the crisis-state heatmap vividly illustrated the formidable challenges faced by energy companies during this turbulent phase, prioritizing the impact of the pandemic over that of the Russia-Ukraine war.

During the COVID-19 pandemic, the cluster analysis revealed the emergence of four distinct clusters, with companies dispersed across each. Each cluster predominantly housed companies from a specific continent and a commonality was observed in the lower PC1 values for a majority of companies. However, as the crisis unfolded, a substantial cluster emerged, signifying a significant transition for most companies. The increase in PC1 values for these companies indicated a pronounced crisis. In contrast, the plot graph for the Russia-Ukraine war, both before and after the conflict, maintained four clusters. The transitions were less pronounced, with clusters remaining relatively stable before and after the war.

The continental responses, eloquently depicted in the R Evolution graph, resonated with the resilience and adaptability of global markets in the face of the pandemic. Notably, during the COVID-19 pandemic, all continents underwent a significant shape transition. When examining the Russia-Ukraine war's impact, the European continent stood out as facing more crises, as evidenced by a slight increase in R compared to other continents. The cluster-based categorization further accentuated the

strategic recalibration of energy companies during both the COVID-19 pandemic and the Russia-Ukraine war.

However, it is crucial to acknowledge the limitations inherent in this study. The reliance on data from Yahoo Finance introduces potential constraints in terms of completeness and accuracy. The simplification of geopolitical events and the exclusion of qualitative insights may limit the depth of our analysis. Recognizing these limitations, the research points toward several promising avenues for future exploration.

Future research endeavors could extend the temporal analysis to capture longer-term trends and assess the sustained impact of geopolitical events on energy companies. Incorporating qualitative methodologies, such as interviews or case studies, could offer a richer understanding of the strategic decisions made by energy companies during crises. Moreover, enhancing the dataset to include energy companies from a more diverse range of geographical regions would contribute to a more comprehensive and globally representative analysis. Integrating macroeconomic indicators into the analysis would provide a holistic perspective, exploring how broader economic trends intersect with geopolitical events to shape energy markets.

Finally, this research, despite its initial focus on the Russia-Ukraine effect, unveils the intricate interplay of global events, positioning the COVID-19 pandemic as a transformative force. By recognizing these dynamics, understanding the study's limitations, and outlining avenues for future research, this work contributes to a nuanced comprehension of the challenges and adaptations within the energy sector amidst complex geopolitical and economic landscapes.

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